

HMI Backend Requirements

1. Data input requirements

The backend shall receive only cleaned, filtered, validated data from WP5.1.

It shall support ingestion of KPI data such as tension, pressure, temperature, speed, machine state, process state, alarms, events, TCP position, TCP orientation, NC/programmed path data, part ID, run ID, timestamp, and contextual metadata.

It shall not own raw DAQ preprocessing, filtering, missing-data correction, or signal validation if those are already owned by WP5.1.

2. Data model requirements

The backend shall define standard schemas for:

- KPI values
- TCP poses
- process states
- process events
- anomaly events
- expected-vs-actual outputs
- root-cause analysis outputs
- geometry mapping outputs
- historical run data

Each record shall include timestamp, run ID, part ID, process state, source system, and data quality/status flag.

3. API requirements

The backend shall provide APIs for:

- dashboard KPI data
- KPI trends
- process timeline

- TCP trajectory
- geometry overlays
- anomaly events
- root-cause analysis outputs
- historical data queries
- expected-vs-actual comparison
- backend health/status

APIs shall support filtering by part ID, run ID, time range, process state, KPI type, anomaly type, and severity.

4. Geometry mapping requirements

The backend shall map:

Timestamp + TCP pose + KPI value



Location on 3D geometry

It shall support KPI overlays, TCP path visualization, anomaly location mapping, process variation mapping, and planned-vs-actual path comparison.

5. Analytics integration requirements

The backend shall integrate outputs from:

- time-series anomaly detection
- behavioral/system identification model
- expected-vs-actual comparison
- root-cause analysis framework

It shall convert these outputs into HMI-ready payloads.

6. Historical query requirements

The backend shall allow users to retrieve past runs, compare runs, inspect anomaly history, replay process timelines, and query KPI behavior over selected time ranges.

7. Performance requirements

The backend shall support responsive dashboard queries, efficient geometry overlay loading, stable API performance, and scalable storage for multiple production runs.

8. Security and access requirements

The backend shall support user roles if required, API authentication, access control, audit logging, and controlled access to process data.

HMI Frontend Requirements

1. General UI requirements

The frontend shall provide a clean HMI interface for process engineers and manufacturing engineers to visualize process intelligence outputs.

It shall include navigation menus, dashboard pages, analysis pages, geometry visualization pages, and historical query pages.

2. Dashboard requirements

The frontend shall display KPI cards, KPI trend charts, process state, run information, part information, machine status, anomaly summary, and process health indicators.

3. KPI visualization requirements

The frontend shall visualize:

- tension trends

- pressure trends
- temperature trends
- speed trends
- KPI variations
- historical KPI behavior
- expected-vs-actual KPI comparison

Users shall be able to filter by time range, run ID, process state, and KPI type.

4. Anomaly visualization requirements

The frontend shall show anomaly events with:

- anomaly type
- timestamp
- severity
- related KPI
- process state
- geometry location
- affected time window

It shall support anomaly timeline, anomaly table, severity filtering, and anomaly drill-down.

5. 3D geometry visualization requirements

The frontend shall display part geometry and allow zoom, pan, rotate, selection, layer control, and overlay visibility control.

It shall display TCP trajectory, programmed path, actual path, KPI heatmaps, process variation overlays, and anomaly markers.

6. Process timeline requirements

The frontend shall show process events and process states over time, including course start/end, pause, resume, alarm, machine stop, state transitions, and run progress.

7. Root-cause analysis view requirements

The frontend shall provide a process investigation view showing related KPIs, anomaly context, possible contributing variables, process-state context, timeline before/after anomaly, and geometry location.

This should be presented as engineering investigation support, not automated diagnosis.

8. Expected-vs-Actual view requirements

The frontend shall show expected process behavior from the behavioral model against actual process behavior.

It shall visualize deviation, deviation magnitude, process envelope, and related KPI trends.

9. Historical analysis requirements

The frontend shall allow selection of previous runs, comparison of runs, replay of process events, anomaly history review, KPI trend comparison, and geometry overlay replay.

10. Usability requirements

The frontend shall be readable, intuitive, responsive, and suitable for engineering review. It shall avoid clutter, allow filtering, support drill-down, and clearly distinguish actual data, expected behavior, anomalies, and root-cause investigation information.

HMI Backend Requirements

ID	Requirement Area	Detailed Backend Requirement
BE-01	Data source boundary	Backend shall receive only cleaned, filtered and validated data from WP5.1. Raw DAQ filtering, noise removal, missing-data correction and signal validation shall remain under WP5.1.
BE-02	DAQ data ingestion	Backend shall ingest KPI/process data such as tension, pressure, temperature, speed, process state, machine state, alarm/event data, run ID, part ID and timestamp.
BE-03	TCP data ingestion	Backend shall ingest actual TCP pose data including timestamp, X/Y/Z position, orientation, machine state and run ID.
BE-04	NC/programmed path ingestion	Backend shall ingest programmed/NC path data so actual TCP trajectory can be compared or overlaid against planned path.
BE-05	Metadata handling	Backend shall maintain metadata such as part ID, run ID, course ID, layer ID, material batch, process phase and geometry reference ID.
BE-06	KPI schema	Backend shall define a standard KPI schema containing KPI name, value, unit, timestamp, source, status flag, process state and quality/data-confidence flag.
BE-07	TCP schema	Backend shall define TCP schema containing timestamp, position, orientation, run ID, process state and associated geometry reference.
BE-08	Event schema	Backend shall define event schema for process start, stop, pause, resume, alarm, anomaly event, operator event and machine-state transition.
BE-09	Geometry schema	Backend shall define geometry reference schema to link part geometry, mesh/CAD ID, coordinate system and mapped process data.
BE-10	Data synchronization	Backend shall synchronize KPI data, TCP pose data, NC path data and event data using timestamp/run ID/process-state references.
BE-11	Data routing	Backend shall route incoming data into visualization-ready streams such as KPI trends, TCP trajectory, event timeline, geometry overlay and historical run datasets.

BE-12	KPI aggregation	Backend shall provide aggregation logic for min, max, average, moving average, time-window summary and run-level KPI summary.
BE-13	Historical storage	Backend shall store historical KPI, TCP, event and geometry-overlay data for later replay and investigation.
BE-14	Run management	Backend shall support retrieval of previous runs by part ID, run ID, date, process state and course/layer information.
BE-15	TCP trajectory processing	Backend shall convert TCP pose sequences into trajectory datasets suitable for 2D/3D visualization and playback.
BE-16	Planned-vs-actual mapping	Backend shall support comparison between programmed NC path and actual TCP path, including spatial deviation and timing deviation.
BE-17	Geometry loading support	Backend shall support loading and referencing part geometry/mesh/CAD representation for visualization purposes.
BE-18	Geometry indexing	Backend shall maintain geometry indexing or lookup mechanism so TCP points and KPI values can be mapped to part regions.
BE-19	KPI-to-geometry mapping	Backend shall map KPI values such as tension, pressure, temperature and speed onto corresponding TCP/geometry locations.
BE-20	Overlay payload generation	Backend shall generate frontend-ready payloads for heatmaps, colored path overlays, markers and region-based visualization.
BE-21	Anomaly integration	Backend shall accept anomaly outputs from the time-series anomaly framework when available and convert them into HMI-ready events.
BE-22	Behavioral model integration	Backend shall accept expected-vs-actual outputs from system identification/behavioral model when available.
BE-23	Root-cause data integration	Backend shall accept root-cause/investigation outputs if available, such as contributing variables, related KPIs and event context.
BE-24	Dashboard API	Backend shall expose APIs for KPI dashboard cards, summary statistics, trend data and process-health information.
BE-25	Trend API	Backend shall expose APIs for time-series KPI trends with filtering by time range, part ID, run ID, process state and KPI type.

BE-26	TCP API	Backend shall expose APIs for actual TCP trajectory, programmed path and planned-vs-actual comparison.
BE-27	Geometry overlay API	Backend shall expose APIs for KPI overlays, anomaly overlays, TCP path overlays and process variation overlays on 3D geometry.
BE-28	Event timeline API	Backend shall expose APIs for process timeline, process-state transitions, alarms, events and run progress.
BE-29	Historical query API	Backend shall support historical queries for previous runs, KPI trends, anomaly history, geometry overlays and process replay.
BE-30	Filter support	Backend APIs shall support filters such as run ID, part ID, time range, KPI type, process state, anomaly severity and geometry region.
BE-31	API response standardization	Backend shall return standardized JSON payloads with timestamp, value, unit, metadata, status and error information.
BE-32	Backend health API	Backend shall provide service health/status endpoint for frontend connectivity checks.
BE-33	Performance	Backend shall support responsive dashboard loading and efficient geometry overlay retrieval for large process datasets.
BE-34	Error handling	Backend shall return clear error messages for missing data, invalid run ID, unavailable geometry, API timeout and malformed payloads.
BE-35	Logging	Backend shall log ingestion status, API requests, data errors, synchronization issues and service failures.
BE-36	Access control	Backend shall support role-based access if required, separating viewer, engineer and admin-level access.
BE-37	Deployment	Backend shall support deployment in the agreed WP5.1/WP5.3 environment with configuration files and environment variables.
BE-38	Documentation	Backend shall provide API documentation, data schema documentation, deployment guide and integration guide.

HMI Frontend Requirements

ID	Requirement Area	Detailed Frontend Requirement
FE-01	General UI layout	Frontend shall provide a clean HMI layout with header, sidebar/navigation menu, main dashboard area and page-level routing.
FE-02	Navigation	Frontend shall provide navigation for dashboard, KPI trends, process timeline, 3D geometry view, anomaly view, root-cause/investigation view and historical run view.
FE-03	Ver1 basic UI	Ver1 shall include basic screens, buttons, navigation, page templates, mock data display and frontend-backend connectivity test.
FE-04	Dashboard landing page	Frontend shall show high-level run information such as part ID, run ID, current process state, active KPIs and latest events.
FE-05	KPI summary cards	Frontend shall display KPI cards showing current value, unit, status, min/max, average and latest timestamp.
FE-06	KPI trend charts	Frontend shall display time-series charts for tension, pressure, temperature, speed and other available KPIs.
FE-07	KPI filtering	Frontend shall allow filtering KPI trends by run ID, time range, KPI type, process state and course/layer if available.
FE-08	KPI comparison	Frontend shall allow side-by-side comparison of multiple KPIs over the same time window.
FE-09	KPI status indication	Frontend shall visually indicate normal, warning and critical KPI status if status limits are available.
FE-10	Process timeline	Frontend shall show process events and state transitions over time, such as run start, course start, pause, alarm, resume and run end.
FE-11	Process-state view	Frontend shall show process-state information such as layup, travel, transition, idle, pause or alarm state if available from data.
FE-12	Event table	Frontend shall provide event table with timestamp, event type, process state, description and severity/status.
FE-13	TCP trajectory view	Frontend shall visualize actual TCP path as a trajectory line using position data.

FE-14	Planned path view	Frontend shall visualize NC/programmed path if available.
FE-15	Planned-vs-actual view	Frontend shall show actual TCP path against programmed/NC path and highlight deviation if computed by backend.
FE-16	3D geometry viewer	Frontend shall provide interactive 3D part/mesh visualization with rotate, zoom, pan and reset view controls.
FE-17	Geometry layer control	Frontend shall allow users to enable/disable layers such as part geometry, TCP path, KPI overlay, anomaly markers and programmed path.
FE-18	KPI heatmap overlay	Frontend shall display KPI values on geometry using heatmap or colored path representation.
FE-19	KPI overlay selection	Frontend shall allow users to select which KPI is overlaid on geometry, such as tension, speed, pressure or temperature.
FE-20	TCP point inspection	Frontend shall allow user to click or hover on TCP/path points to see timestamp, KPI values, process state and position.
FE-21	Anomaly list view	Frontend shall show anomaly events with timestamp, type, severity, related KPI, process state and location if available.
FE-22	Anomaly timeline	Frontend shall show anomalies on the process timeline to indicate when they occurred.
FE-23	Anomaly geometry marker	Frontend shall show anomaly markers on 3D geometry when spatial mapping is available.
FE-24	Anomaly filtering	Frontend shall support filtering by severity, KPI, time range, process state and geometry region.
FE-25	Anomaly drill-down	Frontend shall allow user to open anomaly details showing before/after KPI trends, event context, TCP location and related process state.
FE-26	Expected-vs-actual view	Frontend shall display expected process/KPI behavior and actual behavior when behavioral-model outputs are available.
FE-27	Deviation visualization	Frontend shall show deviation magnitude between expected and actual behavior as trend, table or overlay.
FE-28	Process envelope visualization	Frontend shall display nominal/expected KPI envelope if provided by backend.

FE-29	Root-cause investigation view	Frontend shall provide investigation screen showing related KPIs, process context, contributing variables and event timeline.
FE-30	Root-cause wording	Frontend shall present root-cause output as “engineering investigation support,” not automated diagnosis.
FE-31	Historical run browser	Frontend shall allow user to select historical runs by date, part ID, run ID or process condition.
FE-32	Historical replay	Frontend shall support replay of KPI trends, TCP trajectory, process events and overlays for a selected run.
FE-33	Run comparison	Frontend shall allow comparison of two or more runs for KPI trends and process variation, if backend supports it.
FE-34	Data loading states	Frontend shall show loading, empty, unavailable and error states clearly.
FE-35	API connectivity status	Frontend shall display backend connectivity/service status, especially for Ver1 integration testing.
FE-36	Responsive layout	Frontend shall support dashboard readability across standard engineering workstation screen sizes.
FE-37	User role display	Frontend shall support different views for viewer, engineer and admin roles if access control is required.
FE-38	Export/screenshot	Frontend may support screenshot/export of selected views only if required by stakeholders; otherwise keep optional.
FE-39	UI consistency	Frontend shall follow agreed HMI template, colors, typography, icons, spacing and naming conventions.
FE-40	Usability	Frontend shall minimize clutter, keep important KPIs visible, and allow drill-down only when needed.
FE-41	Performance	Frontend shall render KPI charts and 3D overlays smoothly for realistic dataset sizes.
FE-42	Validation	Frontend shall be validated for navigation, data rendering, API integration, geometry overlay accuracy and user workflow clarity.

Recommended split for release planning

Version	Backend focus	Frontend focus
HMI Ver1	Backend skeleton, mock APIs, health API, mock KPI payloads	UI template, navigation, placeholder screens, mock dashboard, FE-BE integration
HMI Ver2	Real KPI/TCP APIs, geometry mapping APIs, historical query APIs	KPI dashboard, process timeline, TCP path, basic 3D geometry, KPI overlay
HMI Ver3	Anomaly/model/root-cause integration APIs	Anomaly views, expected-vs-actual views, investigation views, mature overlays